

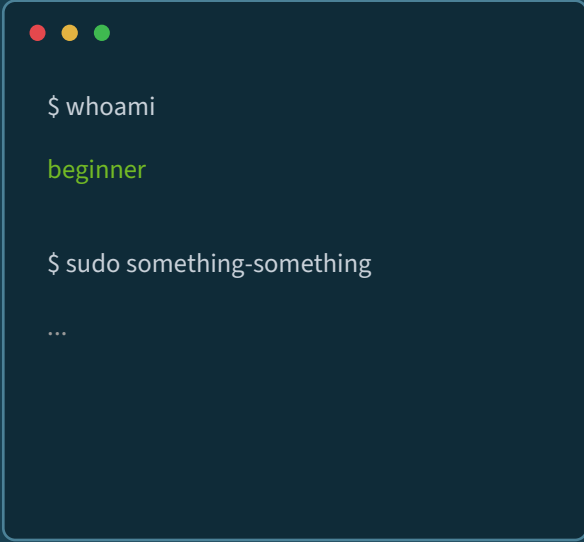
I DON'T KNOW WHAT I JUST **RAN.**

A beginner's guide to not breaking Linux

Common Linux mistakes, terminal gotchas, and learning to think like Linux

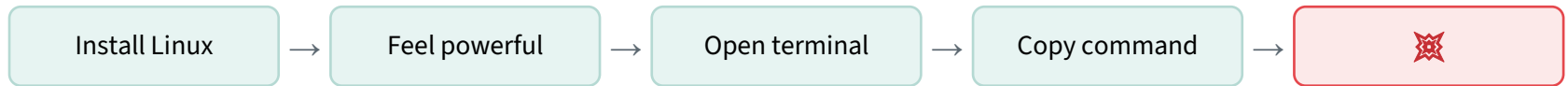
Sk Izaz Ahmed

openSUSE.Asia Summit



```
$ whoami  
beginner  
  
$ sudo something-something  
...
```

The Linux Journey



“Wait... what happened?”

Ask the audience:

“Who has ever run a command without fully knowing what it does?”

The First Lie Linux Tells You



```
$ sudo apt install ...
```

```
✓ works. you feel like a hacker.
```

```
$ sudo ...
```

```
$ sudo ...
```

```
$ sudo ...
```

**“If it doesn't work,
try sudo.”**



*This is one of the fastest ways beginners stop
understanding Linux.*

ACT 2 — THE FIRST MISTAKE

The Internet Said So

“Just run this command.”

```
curl something.com/install.sh | sudo bash
```

Would you run this?

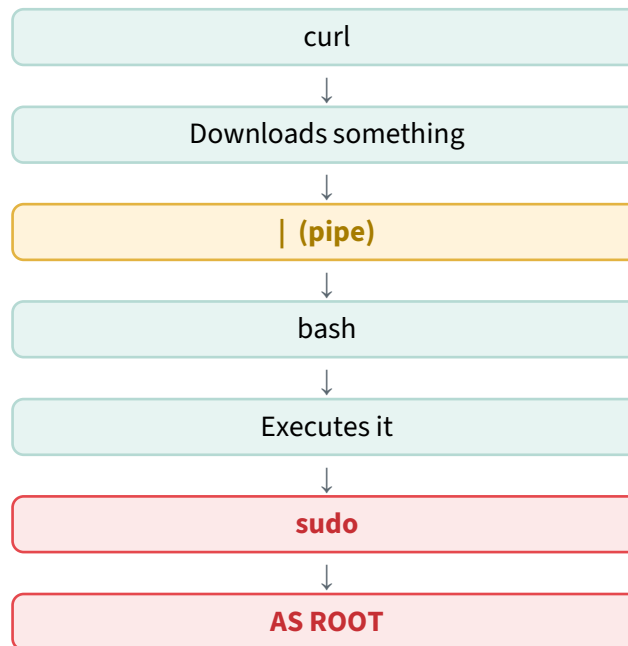
☒ Yes

☐ Maybe

☐ Absolutely not

“What exactly are you trusting here?”

What Did You Actually Run?



Downloading something and executing something are two different actions.

ACT 2 — THE FIRST MISTAKE

The Golden Rule

Read before you run.

Before running a command, ask:

- 1 What does it do?
- 2 What files does it touch?
- 3 Does it need sudo?
- 4 Where did it come from?
- 5 Can I understand the command?

A command being popular doesn't make it safe.

What Does sudo Actually Mean?



```
$ sudo command
```

Means roughly:

“Run this with elevated privileges.”

You are temporarily asking Linux:

**“Trust me
with the keys.”**

Why sudo Feels Like Magic

NORMAL USER

```
/home/shubham/
```

Limited access.
Can only touch what belongs to you.

ROOT

```
/
|— etc |— usr
|— var |— boot
```

Fewer safety barriers.
Access to everything on the system.

Root isn't “better user mode.” Root has fewer safety barriers.

ACT 3 — “SUDO FIXED IT”

The rm Moment

```
rm -rf something
```

rm

→ remove

-r

→ recursively

-f

→ force

Three tiny characters can have very large consequences.

(We won't show the real-world destructive versions involving / — you know the ones.)

Gotcha #1

Linux does exactly what you ask.

Sometimes that's the problem.

```
$ mv report.txt reports/
```

No AI.
No guessing.
No “Are you sure?”

You expected:

“Move my report.”

Linux understood:

“Move this file to that exact location.”

Linux Is Case-Sensitive

Documents

documents

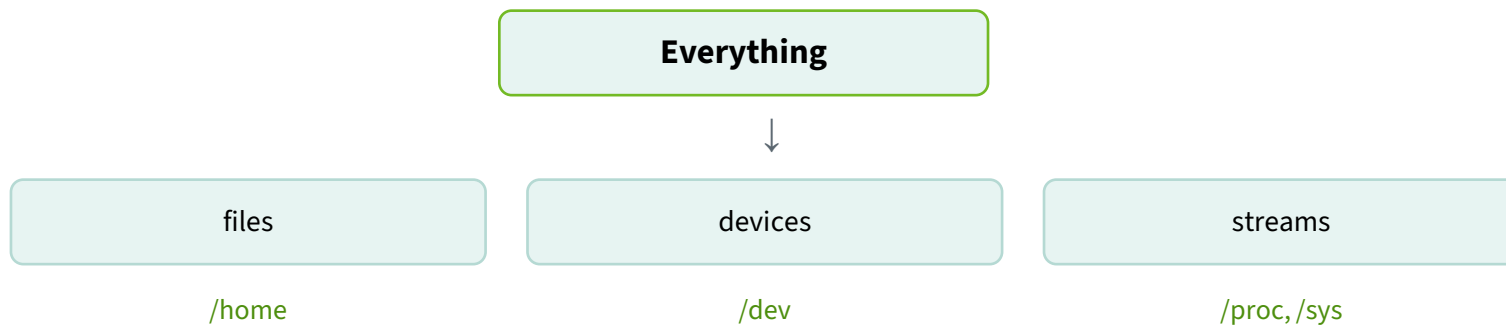
DOCUMENTS

These can all be different files.

Why?

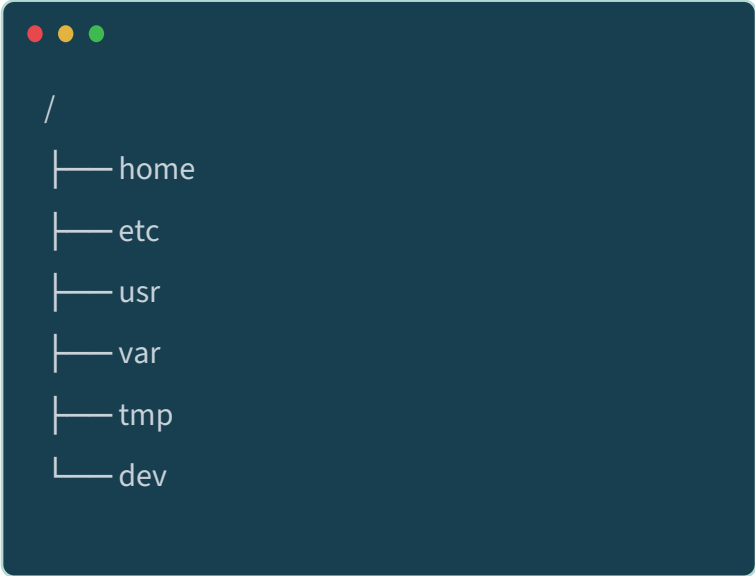
Because Linux treats names literally — byte for byte.

Everything Is a File



Linux has a surprisingly elegant way of making very different things look similar to programs.

The / Problem



```
/
├── home
├── etc
├── usr
├── var
├── tmp
└── dev
```

Linux doesn't have C:\ and D:\ like Windows.

Everything starts from:

/

ACT 4 — LINUX DOESN'T THINK LIKE YOU

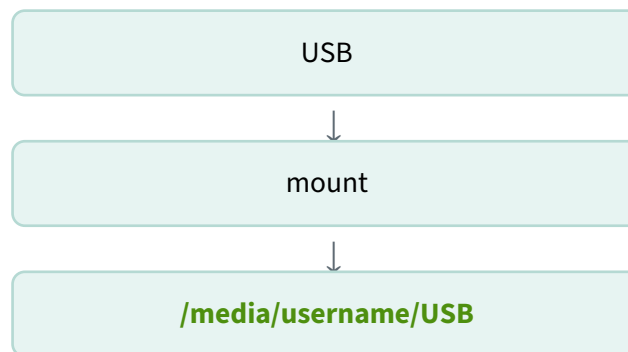
Where Did My USB Go?

Beginner:

"I plugged in my USB. Where is drive E?"

Linux:

"Drive letters? We don't do that here."



Linux Says: Permission Denied



```
$ ./script.sh
```

```
Permission denied
```

Beginner: "Linux is broken."

Linux: "No. You're not allowed." 😏

Read / Write / Execute



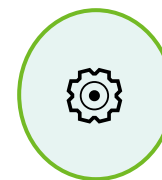
r

read



w

write



x

execute



Linux asks: Who are you, and what are you allowed to do?

Gotcha #2

“It works with sudo.”

That doesn't necessarily mean:

“sudo fixed the problem.”

It may mean:

“You bypassed the permission boundary.”

“I'll Just Download the App”

Windows → download .exe

Linux → 🤖

Meet the Package Manager

- Where software comes from
- What version you have
- Dependencies
- Updates
- What files belong to the package

Linux Software ≠ Random Downloads

Random installer

????

install

Repository → Package Manager

Repository

Package manager

Dependencies

Install → Updates

```
$ zypper install package
```

Package Manager Gotcha

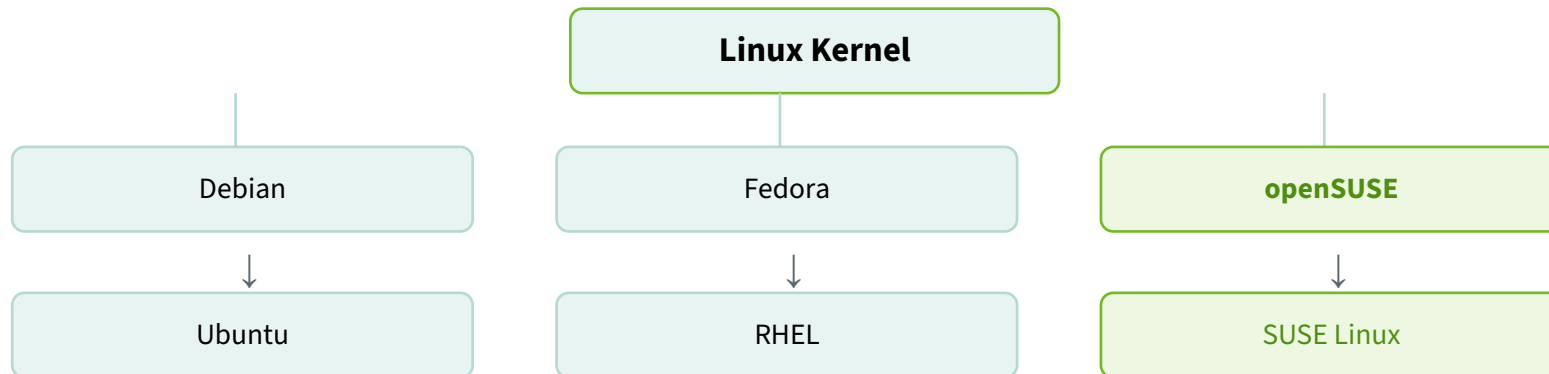
“Why doesn't `sudo apt install ...` work on openSUSE?”

Because Linux is not one operating system.

Linux is the kernel.

Different distributions build different userlands, tools, package ecosystems, defaults, and experiences around it.

The Linux Family



Linux is more like a family than a single product.

Why Do People Choose Linux?



Control



Customisation



Transparency



Powerful tooling



Open source



Developer-friendly

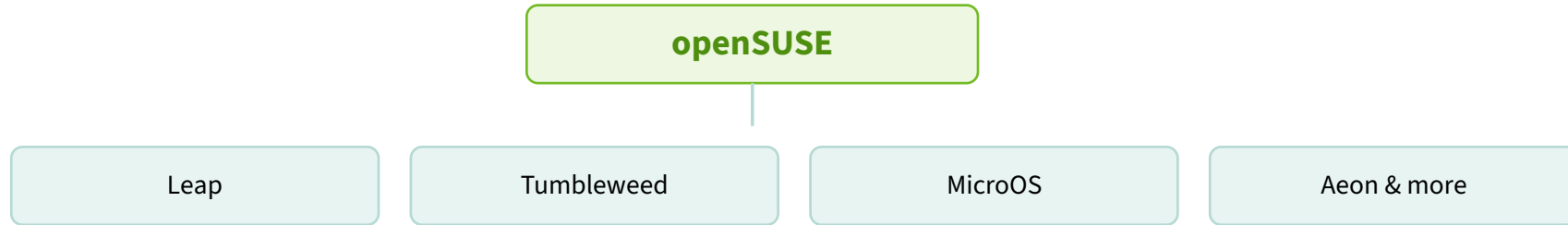


Runs almost everywhere

“You don't have to use Linux because you're a developer.”

“You use it when you want more control over your computer.”

Okay, But Why openSUSE?



Focus on the philosophy rather than overwhelming people with distro details.

Gotcha #3

“>” is NOT “>>”

```
echo hello > file.txt
```

“>” overwrite

```
echo hello >> file.txt
```

“>>” append

Which one would you use in a log file?

Gotcha #4

Spaces matter

```
cd My Folder
```

isn't
the
same
as

```
cd "My Folder"
```

```
cd My\ Folder
```

The shell parses your command before the program ever sees it.

That's a powerful mental model.

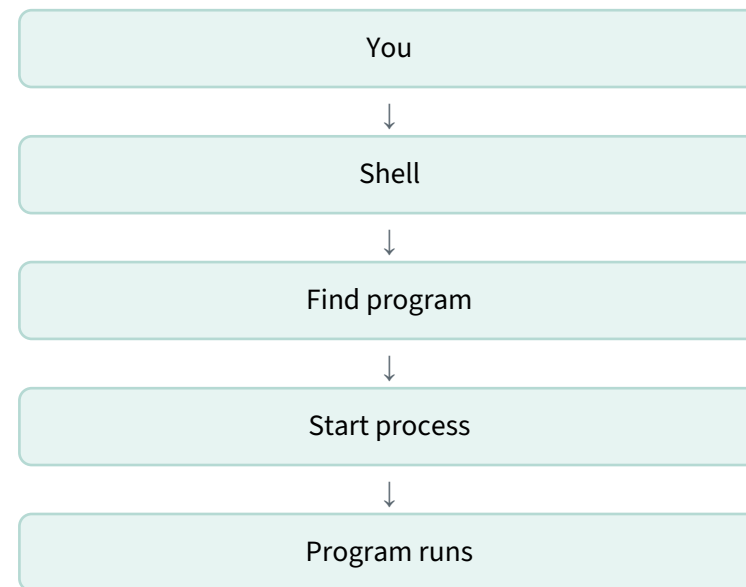
Gotcha #5

The terminal isn't magic.



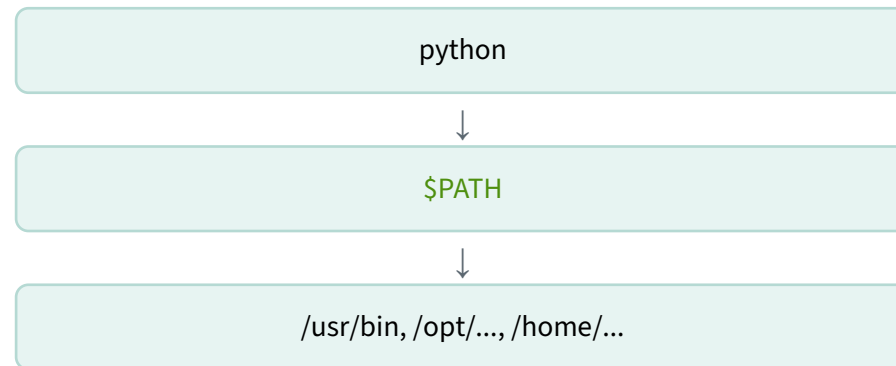
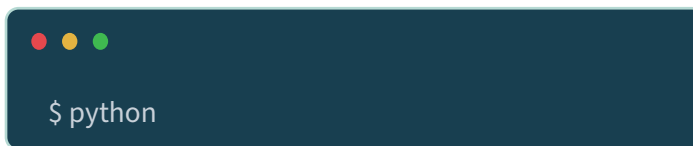
```
$ python app.py
```

Once students understand this, commands become less mysterious.



Gotcha #6

PATH is not a folder



The classic: “It works in one terminal but not another.”

Don't Panic.



Beginner



Experienced user



- Don't immediately run another command.
- Actually read the error.
- What changed?
- Make one change.
- Did it actually fix the problem?

The Most Powerful Debugging Question

“What changed?”

“It worked yesterday.”

Package update?

Config?

Environment variable?

Permissions?

Network?

My code?

Much more useful than “Linux randomly broke.”

You're Sitting on a Superpower



Desktops



Cloud servers



Android



Spacecraft



Networking



AI infra



Gaming



The internet



Embedded

Your laptop is running the same kernel family used across an enormous range of systems.

Why Developers Love Linux

ssh

git

curl

grep

sed

awk

find

tar

systemctl

docker

python

gcc

The terminal isn't the point.

The composability is.

One tool does one thing. Tools can be combined.

Linux Isn't the Only Choice

If you care about...	You might choose...
Ecosystem / desktop apps	Windows
Apple hardware integration	macOS
Control / servers / dev tooling	Linux
Gaming	Windows + Linux improving
Learning systems	Linux
Privacy / control	Linux
Enterprise workloads	Depends

The goal isn't to convince everyone to use Linux.

It's to understand why people choose it.

Stop Memorising Commands

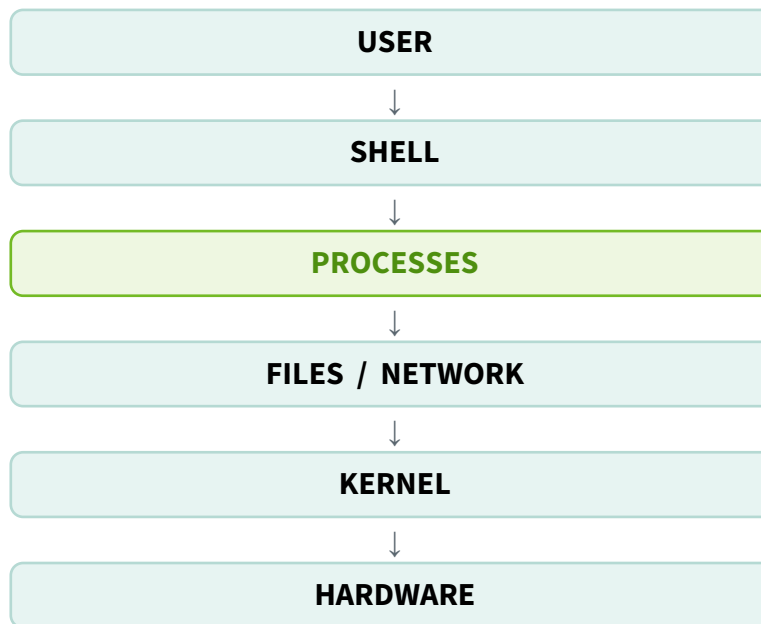
Memorise 100 commands

↓ instead ↓

Understand 10 concepts,
learn commands when needed

- Processes
- Files
- Permissions
- Pipes
- Redirection
- Packages
- Environment
- Networking

Your Linux Mental Model



Once you understand the model, commands stop looking random.

Spot the Mistake

```
sudo curl https://example.com/script.sh | bash
```

What's wrong?

- Why sudo?
- Why pipe directly to bash?
- Do we trust the source?
- What does the script do?
- Why not inspect it first?

Final Challenge

```
$ command  
command not found
```

- Read the error
- Check spelling
- Check PATH
- Check whether it's installed
- Check documentation
- Understand before installing random things

What do you do next?

NOT: sudo command 🤪

The Linux Beginner's Rulebook

- 01** Don't copy commands blindly.
- 02** Don't use sudo just because something failed.
- 03** Read the error message.
- 04** Change one thing at a time.
- 05** Understand the concept, not just the command.

Linux isn't about knowing commands.

It's about knowing what to ask when something goes wrong.

“I don't fully understand what I just ran...”

“I don't know yet.

But I know how to find out.”

Welcome to Linux.

- Break things.
- Read the error.
- Understand why.
- Fix them.
- Learn something.

Thank You

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